

EQUALITY.

RESERVES#ACTIVATION_MFRR_STANDARD_SA

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Change history

Version	Author	Date	Description
1.0	Don Boogerd	03 Jun 2026	Initial version
1.1	Xander Pikaar	13 Jul 2026	Technical details added
1.2	Don Boogerd	16 Jul 2026	IM Reviewed with Transparency Team and approved
1.3	Xander Pikaar	16 Jul 2026	XSD-version changed

Review information

Who	Reviewed (date)
Erik Baaijens	24 Jun 2026
Xander Pikaar	Co-author
Sander Maas	14 Jul 2026
Sandjai Ramasray	14 Jul 2026
Nico Klaver	16 Jul 2026

Intro

MARI staat voor **Manually Activated Reserves Initiative**.

In de Europese energiemarkt is MARI het gezamenlijke platform waarop Transmission System Operators (TSO's), **TenneT**, automatisch de goedkoopste beschikbare **mFRR (manual Frequency Restoration Reserve)**-energie kunnen uitwisselen om het elektriciteitsnet in balans te houden.

Scheduled Activation (SA) is een standaard MARI-proces waarbij mFRR-biedingen periodiek worden geactiveerd om verwachte onbalansen in het elektriciteitsnet op te lossen. Het MARI-platform voert hiervoor elke 15 minuten een optimalisatie uit.

Functional Information

Description	EQUALITY.RESERVES#ACTIVATION_MFRR_STANDARD_SA
Content	For mFRR Standard product (MARI) energy bids can be provided and activated with Scheduled Activation or Direct Activation. In this delivery the Scheduled Activations are provided per market Party (BSP) per ISP.
Measure Unit	MAW
Resolution (Program Time Unit)	PT15M
Time series	1 ..
Document type	Activation_MarketDocument
DDA deadline	n/a

Frequency	Provided in case of activation, within minutes after the end of the ISP of the activation
Period	15 Minutes

Source information
- Module ASM - Module ASM

The delivery is used by the following publications
- EDP_74mFRRsa_Standard Aggregated balancing energy bids - EDP_74mFRRsa_Standard Aggregated balancing energy bids

Functional requirements			
Requirement	Described in IM	Built	Approved

Functional Notes	

Technical Information

XSD	
Document	Activation_MarketDocument
XSD	iec62325-451-7-activationdocument_v6_3
Version	6.3
Namespace	iec62325.351:tc57wg16:451-7:activationdocument:6:3
File	 iec62325-451-7-activationdocument_v6_3.xsd

Source system

Queue name	...
Queue table	...
Publishing procedure	...

Target system (delphi)	
Queue name	i_aq_evt_delphi
Queue table	jms_i_evt_messages
Dequeue Job	dequeue_i_aq_evt_delphi
Parser Package	rcn_parse_activation_md
Global temporary table (GTT)	activation_md_gtt activation_md_timeseries_gtt activation_md_period_gtt
Processing procedure (RCN package)	rcn_power_activations.process_activation_mfrr_sa
Check Package	...
Table space	bln_data, bln_index
Target tables	bln_power_activations. bln_pan_values
Used MRD tables	...

Parameter values	
Message_type	equality.event translated to EQUALITY.RESERVES
Subtype	ACTIVATION_MFRR_STANDARD_SA
legal_owner	TTN
lamu_source	EQUALITY

Table structure

bln_power_activations		
P	* id	NUMBER (10)
U	* power_type	VARCHAR2 (100)
U	* mpt_code	VARCHAR2 (100)
U	* mpt_code_type	VARCHAR2 (100)
U	* mpt_object_type	VARCHAR2 (100)
	* tvalidity_utc_from	TIMESTAMP
	* tvalidity_loc_from	TIMESTAMP
	* lamu_user	VARCHAR2 (100)
	* lamu_pcs_id	NUMBER (10)
	* lamu_source	VARCHAR2 (100)
bln_pan_pk(id)		
bln_pan_uk(power_type, mpt_code, mpt_code_type, mpt_object_type)		
bln_pan_pk_idx(id)		
bln_pan_uk_idx(power_type, mpt_code, mpt_code_type, mpt_object_type)		



bln_pan_values		
P	* bvalidity_utc_from	DATE
	* bvalidity_utc_to	DATE
PF	* pan_id	NUMBER (10)
P	* bid_id	VARCHAR2 (100)
P	* direction	VARCHAR2 (10)
P	* order_marketdocument_mrid	VARCHAR2 (100)
	* order_marketdocument_version	NUMBER (10)
	* ptu_date_loc	DATE
	* ptu	NUMBER (5)
	* ptu_resolution	VARCHAR2 (10)
	* power_unit	VARCHAR2 (10)
	* power	NUMBER (13,3)
	* org_power	NUMBER (13,3)
	* isp_bvalidity_utc_from	DATE
	* isp_bvalidity_utc_to	DATE
	* tvalidity_utc_from	TIMESTAMP
	* tvalidity_loc_from	TIMESTAMP
	* lamu_user	VARCHAR2 (100)
	* lamu_pcs_id	NUMBER (10)
	* lamu_source	VARCHAR2 (100)
panvle_pk(bvalidity_utc_from, pan_id, order_marketdocument_mrid, bid_id, direction)		
bln_panvle_pan_fk(pan_id)		
bln_panvle_pk_idx(bvalidity_utc_from, pan_id, order_marketdocument_mrid, bid_id, direction)		

Technical Notes

Message example

Interface mapping

Comment:	...
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Default/Translation		Database	XML			Unhappy Flow						
		Table.column	Element	Level	Occurrence	Check Transformation/Default items are mandatory and must be seen as the "Happy Flow". Any deviation will trigger an error message and a functional activity.	Error message (ENTSO-e Code)	Functional Activity				
			Activation_Market Document	1	1							
Unique identification of the Activation_MarketDocument Example: 550e8400-e29b-41d4-a716-446655440000_8712423009332_202509232200		pcs_rcn_receptions. document_mrid	mRID	2	1							
A unique version within the mRID		pcs_rcn_receptions. document_version	revisionNumber	2	1							
Translate <table><tr><th>Code</th><th>Translation</th></tr><tr><td>B23</td><td>OFFERS_TO_BE_ACTIVATED</td></tr></table>		Code	Translation	B23	OFFERS_TO_BE_ACTIVATED	pcs_rcn_receptions. document_type	type	2	1			
Code	Translation											
B23	OFFERS_TO_BE_ACTIVATED											
Translate <table><tr><th>Code</th><th>Translation</th></tr><tr><td>A60</td><td>STANDARD_MANUAL_FREQUENCY_RESTORATION_RESERVE_SCHEDULED_ACTIVATION Opmerking: SCHEDULED_ACTIVATION_MFRR is de echte vertaling van ENTSO-e voor dit process_type. Om onderscheid te maken tussen mFRRsa voor MARI en "normaal" mFRRsa moeten we hier een aparte vertaling maken.</td></tr></table>		Code	Translation	A60	STANDARD_MANUAL_FREQUENCY_RESTORATION_RESERVE_SCHEDULED_ACTIVATION Opmerking: SCHEDULED_ACTIVATION_MFRR is de echte vertaling van ENTSO-e voor dit process_type. Om onderscheid te maken tussen mFRRsa voor MARI en "normaal" mFRRsa moeten we hier een aparte vertaling maken.	pcs_rcn_receptions. process_type bln_power_activations. power_type	process. processType	2	1			
Code	Translation											
A60	STANDARD_MANUAL_FREQUENCY_RESTORATION_RESERVE_SCHEDULED_ACTIVATION Opmerking: SCHEDULED_ACTIVATION_MFRR is de echte vertaling van ENTSO-e voor dit process_type. Om onderscheid te maken tussen mFRRsa voor MARI en "normaal" mFRRsa moeten we hier een aparte vertaling maken.											
Expect <table><tr><th>Code</th><th>Meaning</th></tr><tr><td>8716867999983</td><td>TenneT NL</td></tr></table>		Code	Meaning	8716867999983	TenneT NL	pcs_rcn_receptions. document_sender_mrid	sender_MarketParticipant.mRID	2	1			
Code	Meaning											
8716867999983	TenneT NL											
Expect <table><tr><th>Code</th><th>Translation</th></tr><tr><td>A10</td><td>EAN-Code</td></tr></table>		Code	Translation	A10	EAN-Code	Not used	sender_MarketParticipant.mRID /@coding/Scheme		1			
Code	Translation											
A10	EAN-Code											
Expect <table><tr><th>Code</th><th>Meaning</th></tr><tr><td>A04</td><td>System operator</td></tr></table>		Code	Meaning	A04	System operator	Not used	sender_MarketParticipant.marketRole.type	2	1			
Code	Meaning											
A04	System operator											
PartyID_String - EAN Code of BSP		Not used	receiver_MarketParticipant.mRID	2	1							
Expect <table><tr><th>Code</th><th>Meaning</th></tr><tr><td>A10</td><td>EAN-Code</td></tr></table>		Code	Meaning	A10	EAN-Code	Not used	receiver_MarketParticipant.mRID /@codingScheme		1			
Code	Meaning											
A10	EAN-Code											
Expect <table><tr><th>Code</th><th>Meaning</th></tr><tr><td>A46</td><td>Balancing Service Provider</td></tr></table>		Code	Meaning	A46	Balancing Service Provider	Not used	receiver_MarketParticipant.Role.type	2	1			
Code	Meaning											
A46	Balancing Service Provider											
Format 'yyyy-mm-ddThh24:mi:ssZ"		Not used	createdDateTime	2	1							
			activation_Time_Period.timeInterval	2	1							
Format 'yyyy-mm-ddThh24:mi:Z"		Not used	start	3	1							
Format 'yyyy-mm-ddThh24:mi:Z"		Not used	end	3	1							

Expect		Not used	domain.mRID	2	1							
<table><tr><th>Code</th><th>Meaning</th></tr><tr><td>10YNL-----L</td><td>The Netherlands</td></tr></table>	Code	Meaning	10YNL-----L	The Netherlands								
Code	Meaning											
10YNL-----L	The Netherlands											
Translate		Not used	domain.mRID /@codingScheme		1							
<table><tr><th>Code</th><th>Translation</th></tr><tr><td>A01</td><td>EIC_CODE</td></tr></table>	Code	Translation	A01	EIC_CODE								
Code	Translation											
A01	EIC_CODE											
PartyID_String - EAN of BSP		Not used	subject_MarketParticipant.mRID	2	1							
Expect		Not used	subject_MarketParticipant.mRID /@codingScheme		1							
<table><tr><th>Code</th><th>Meaning</th></tr><tr><td>A10</td><td>EAN-Code</td></tr></table>	Code	Meaning	A10	EAN-Code								
Code	Meaning											
A10	EAN-Code											
Expect		Not used	subject_MarketParticipant.marketRole.type	2	1							
<table><tr><th>Code</th><th>Translation</th></tr><tr><td>A46</td><td>Balancing Service Provider</td></tr></table>	Code	Translation	A46	Balancing Service Provider								
Code	Translation											
A46	Balancing Service Provider											
Unique identification of the Energy Bid Document Example: 4284eeb6-8c66-4e4b-963e-5a5309f02e09		bln_pan_values. order_marketdocument_mrid	order_MarketDocument.mRID	2	1							
The identification of the version that distinguishes one evolution of a document from another.		bln_pan_values. order_marketdocument_version	order_MarketDocument. revisionNumber	2	1							
			TimeSeries	2	1..							
A unique identification of the time series.		bln_pan_values.bid_id	mRID	3	1							
PartyID_String - EAN of BSP		bln_power_activations. mpt_code bln_power_activations. mpt_object_type = 'BALANCING_SERVICE_PROVIDER'	resourceProvider_MarketParticipant.mRID	3	1							
Expect		bln_power_activations. mpt_code_type	resourceProvider_MarketParticipant.mRID /@codingScheme		1							
<table><tr><th>Code</th><th>Meaning</th></tr><tr><td>A10</td><td>EAN-Code</td></tr></table>	Code	Meaning	A10	EAN-Code								
Code	Meaning											
A10	EAN-Code											
Expect		not used	businessType	3	1							
<table><tr><th>Code</th><th>Meaning</th></tr><tr><td>B74</td><td>Offer</td></tr></table>	Code	Meaning	B74	Offer								
Code	Meaning											
B74	Offer											
Translate		Not used	acquiring_Domain.mRID	3	1							
<table><tr><th>Code</th><th>Translation</th></tr><tr><td>10YNL-----L</td><td>Bidding_zone of TenneT-NL</td></tr></table>	Code	Translation	10YNL-----L	Bidding_zone of TenneT-NL								
Code	Translation											
10YNL-----L	Bidding_zone of TenneT-NL											
Expect		Not used	acquiring_Domain.mRID /@codingScheme		1							
<table><tr><th>Code</th><th>Meaning</th></tr><tr><td>A01</td><td>EIC_CODE</td></tr></table>	Code	Meaning	A01	EIC_CODE								
Code	Meaning											
A01	EIC_CODE											
Expect		Not used	connecting_Domain.mRID	3	1							
<table><tr><th>Code</th><th>Meaning</th></tr><tr><td>10YNL-----L</td><td>Bidding_zone of TenneT-NL</td></tr></table>	Code	Meaning	10YNL-----L	Bidding_zone of TenneT-NL								
Code	Meaning											
10YNL-----L	Bidding_zone of TenneT-NL											
Expect		Not used	connecting_Domain.mRID /@codingScheme		1							
<table><tr><th>Code</th><th>Meaning</th></tr><tr><td>A01</td><td>EIC</td></tr></table>	Code	Meaning	A01	EIC								
Code	Meaning											
A01	EIC											

Expect		bln_pan_values. power_unit	measurement_Unit. name	3	1									
<table><tr><th>Code</th><th>Meaning</th></tr><tr><td>MAW</td><td>Megawatt</td></tr></table>	Code	Meaning	MAW	Megawatt										
Code	Meaning													
MAW	Megawatt													
Translate		bln_pan_values.direction	flowDirection. direction	3	1									
<table><tr><th>Code</th><th>Translation</th></tr><tr><td>A01</td><td>UP</td></tr><tr><td>A02</td><td>DOWN</td></tr></table>	Code	Translation	A01	UP	A02	DOWN								
Code	Translation													
A01	UP													
A02	DOWN													
Translate		not used	marketObjectStatus. .status	3	1									
<table><tr><th>Code</th><th>Meaning</th></tr><tr><td>A07</td><td>Activated</td></tr></table>	Code	Meaning	A07	Activated										
Code	Meaning													
A07	Activated													
			Period	3	1									
			timeInterval	4	1									
Format 'yyyy-mm-ddThh24:mi:Z'		Use to determine - bln_pan_values. bvalidity_utc_from - bln_pan_values. isp_bvalidity_utc_from - ptu - ptu_date_loc	start	5	1									
Format 'yyyy-mm-ddThh24:mi:Z'		Use to determine - bln_pan_values. bvalidity_utc_to - bln_pan_values. isp_bvalidity_utc_to	end	5	1									
PT15M		bln_pan_values. ptu_resolution	resolution	4	1									
			Point	4	1									
Always 1.		Even though the value is always 1, use the default calculation to determine - bln_pan_values. bvalidity_utc_from - bln_pan_values. isp_bvalidity_utc_from - ptu - bln_pan_values. bvalidity_utc_to - bln_pan_values. isp_bvalidity_utc_to	position	5	1									
Quantity offered with 1 MW precision. Always positive values (no sign).)		bln_pan_values. org_power bln_pan_values.power (same value)	quantity	5	1									

Integration

00. System	
01. Type	Reception
02. Integration	
03. Sender organization	
04. Sender application	
05. Receiver organization	
06. Receiver application	
07. Document type	

08. Component	
09. Module	

UAT status

System	Delphi				
Jira	key	summary	status	customfield_10004	customfield_13905
	JQL and issue key arguments for this macro require at least one Jira application link to be configured				
UAT					
PAT					
GoLive					